

JKBOSE



2026 CLASS 11TH

BASIC MATH AND CALCULUS

LECTURE-7

PHYSICS

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Topics To Be Covered

- 1** Revision
- 2** Differentiation
- 3** Integration
- 4** Board and NEET PYQs
- 5** ..

ONLY THROUGH TIME,
TIME IS CONQUERED

Recap From 1st Lec.

Maths

- Maths Bohat Asaan hai
- number
- + - x ÷
- Fraction
- Angle / Power
- Trigonometry
- Graph / Equation
- $\frac{dy}{dx}$ $\int y dx$

Fraction

- $\frac{1}{2} = 0.5$
- $\frac{1}{3} = 0.33$
- $\frac{1}{4} = 0.25$
- $\frac{1}{10} = 0.1$
- $\frac{1}{100} = 0.01$
- $5.42 \times 100 = 542$

Power

- $2^2 = 2 \times 2$
- $x^0 = 1$
- $x^a \cdot x^b = x^{a+b}$
- $(x^a)^b = x^{a \times b}$

$$\frac{1}{0} = \infty$$

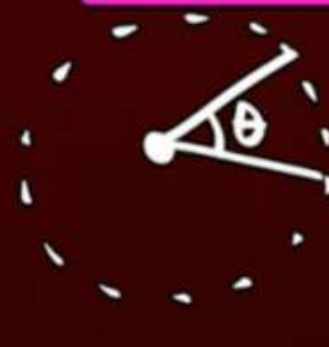
$$\frac{1}{\infty} = 0$$

$$\sqrt{2} = 2^{\frac{1}{2}}$$

$$\sqrt{2} = 1.4$$

$$\sqrt[3]{3} = 1.7$$

Angle θ



$$\theta = \frac{\text{Arc}}{\text{Radius}}$$

0 Rad.

$$\rightsquigarrow 180^\circ = \pi \text{ rad}$$

$$1^\circ = \frac{\pi}{180} \text{ rad}$$

$$1 \text{ rad} = \frac{180^\circ}{\pi}$$

Recap From 2nd Lec.

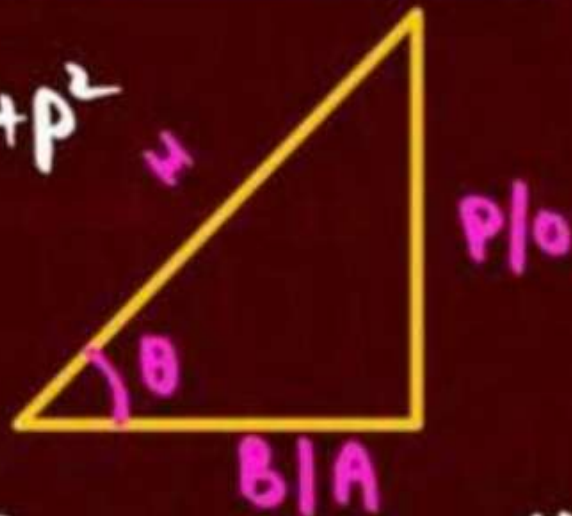
Trigonometry Table

	0°	30°	45°	60°	90°	120°	135°	150°	180°
Angle	0°	30°	45°	60°	90°	120°	135°	150°	180°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{1}{\sqrt{2}}$	$-\frac{\sqrt{3}}{2}$	-1
$\tan \theta$	$\frac{0}{1} = 0$	$\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	$\frac{1}{0} = \infty$	$-\sqrt{3}$	-1	$-\frac{1}{\sqrt{3}}$	0

Trigonometry

Special Δ

$H^2 = B^2 + P^2$



$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$

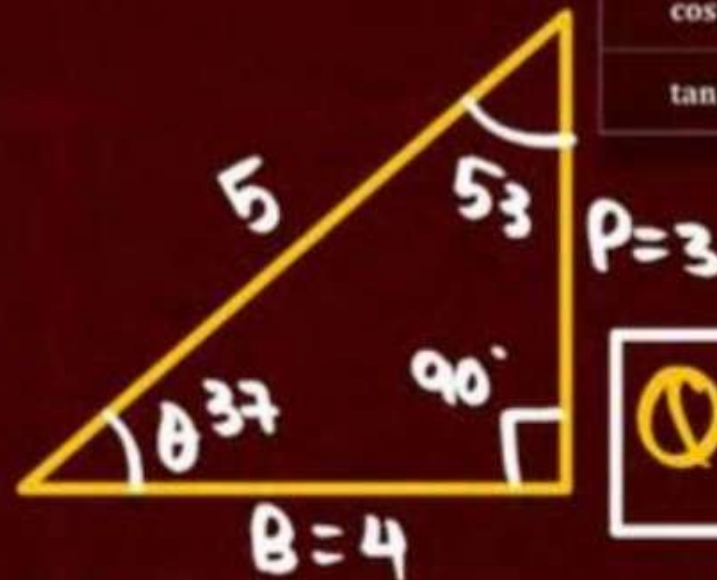
$\sec \theta = \frac{1}{\cos \theta}$

$\cot \theta = \frac{1}{\tan \theta}$

$\sin^2 \theta + \cos^2 \theta = 1$

$\sin \theta = \frac{P}{H}$
 $\cos \theta = \frac{B}{H}$
 $\tan \theta = \frac{P}{B}$

$\operatorname{cosec} \theta = \frac{H}{P}$
 $\sec \theta = \frac{H}{B}$
 $\cot \theta = \frac{B}{P}$



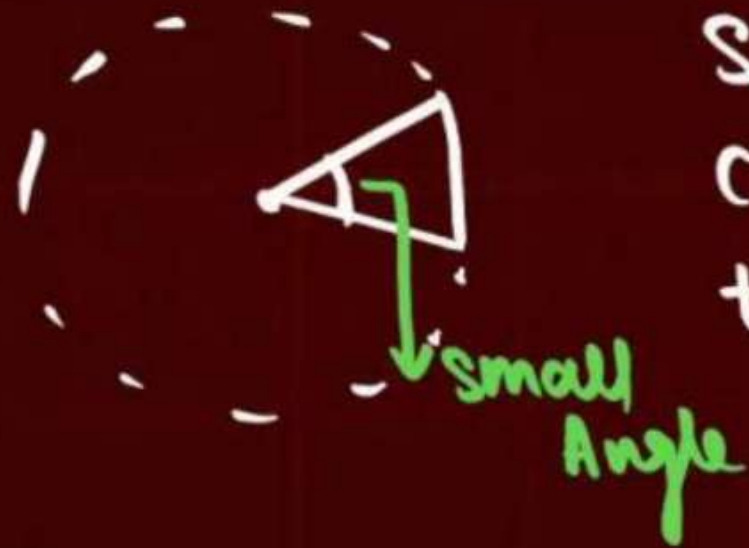
Quadrant System



Recap From 3rd Lec.



Small Angle Approx.



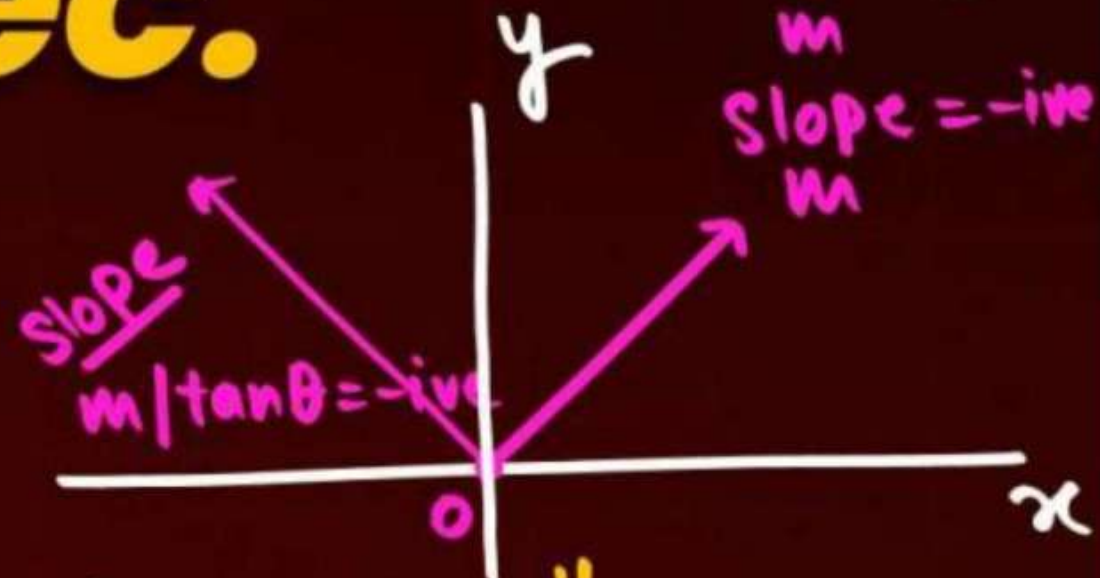
$$\begin{aligned}\sin \theta &= \theta \\ \cos \theta &= 1 \\ \tan \theta &= \theta\end{aligned}$$

Equation

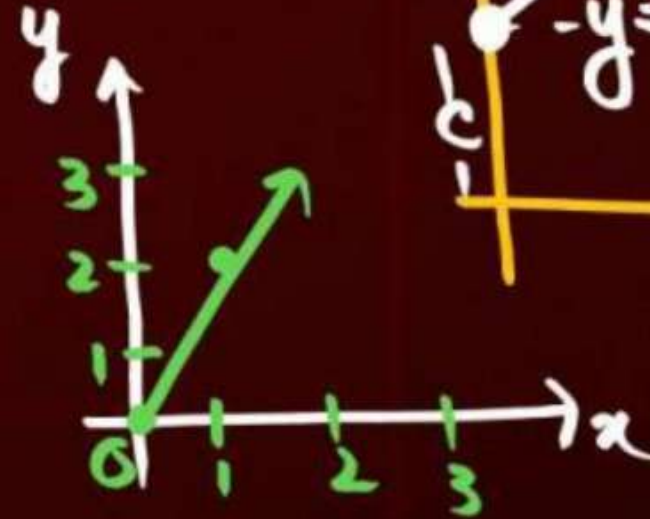
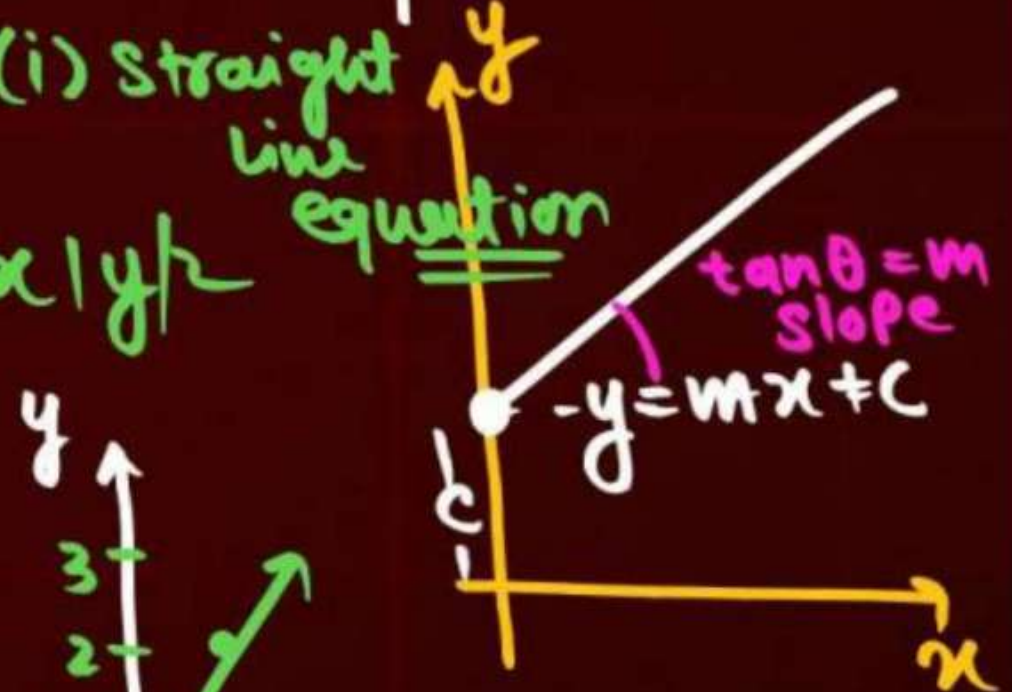
English Statement

Slope $\rightarrow$ $y = \frac{y}{x}x = x \div x | y/x$

x	0	1	2
y	0	2	4



(i) Straight line equation



Recap From 4th Lec. 3rd



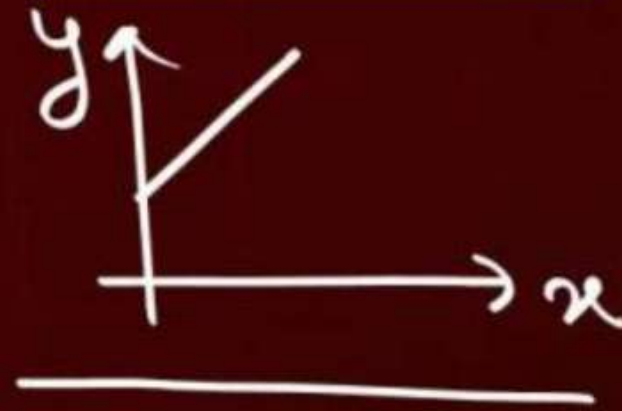
Equation

$$\underline{y = mx + c}$$

$$y = x^2$$

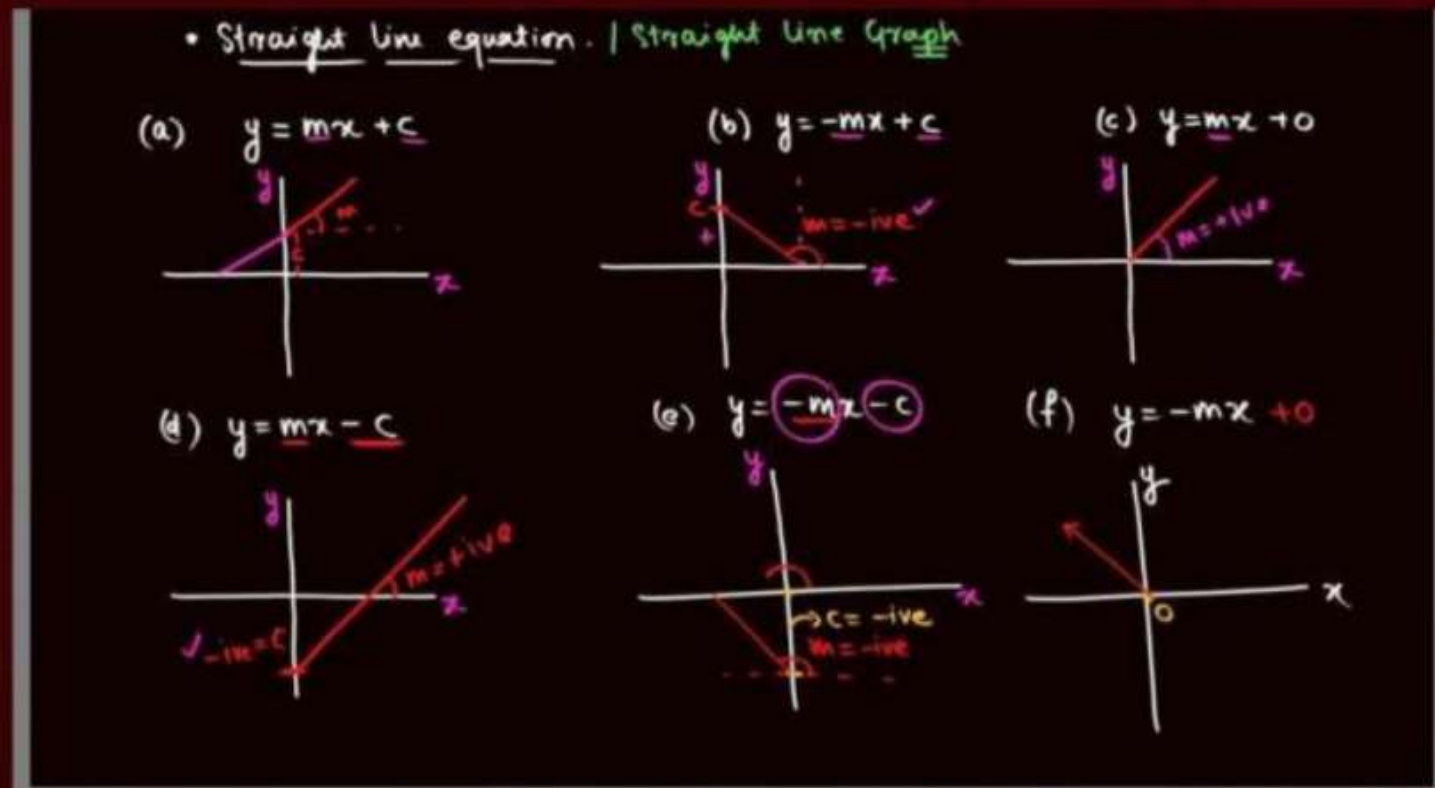
→ Table →

Graph



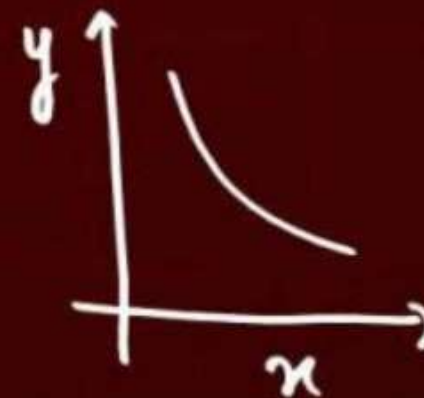
Recap From 5th Lec.

① $y = \underline{mx} + c$ (Important)



② $y = x^2$

③ $y = \frac{1}{x}$



④ $x^2 + y^2 = R^2$

Recap From 6th Lec.

Differentiation

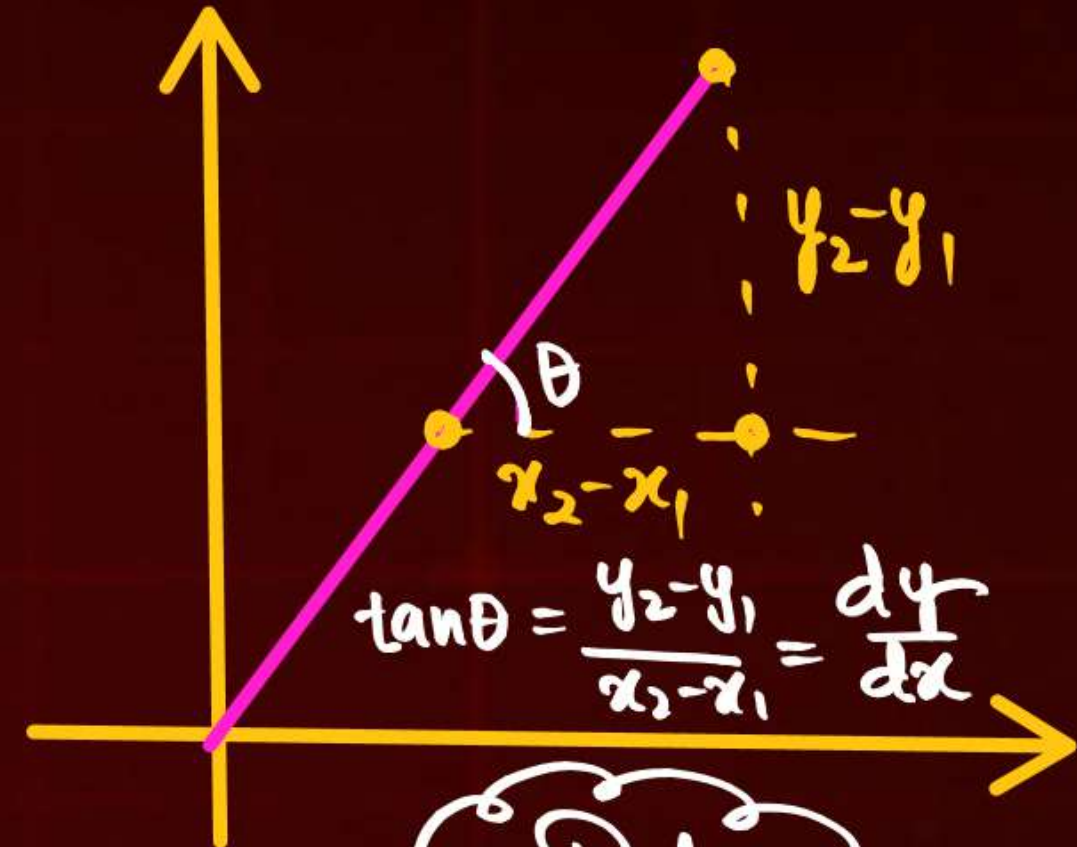
$$y = 2x$$

x	1	→ 2	→ 3	→ 4
y	2	→ 4	→ 6	→ 8

$$y = 2x$$

↓
Slope

differentiation $\left(\frac{dy}{dx}\right)$



$$\text{differentiation} = \frac{\text{Small change in } y}{\text{Small change in } x} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{dy}{dx} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{4 - 2}{2 - 1} = \frac{2}{1} = 2$$

Rules



Isaac Newton
(1642 – 1727)

Isaac Newton was born in Woolsthorpe near Grantham, England. He is generally regarded as the most original and influential theorist in the history of science. He was born in a poor farming family. But he was not good at farming. He was sent to study at Cambridge University in 1661. In 1665 a plague broke

out in Cambridge and so Newton took a year off. It was during this year that the incident of the apple falling on him is said to have occurred. This incident prompted Newton to explore the possibility of connecting gravity with the force that kept the moon in its orbit. This led him to the universal law of gravitation. It is remarkable that many great scientists before him knew of gravity but failed to realise it.

Newton formulated the well-known laws of motion. He worked on theories of light and colour. He designed an astronomical telescope to carry out astronomical observations. Newton was also a great mathematician. He invented a new branch of mathematics, called calculus. He used it to prove that for objects outside a sphere of uniform density, the sphere behaves as if the whole of its mass is concentrated at its centre. Newton transformed the structure of physical science with his three laws of motion and the universal law of gravitation. As the keystone of the scientific revolution of the seventeenth century, Newton's work combined the contributions of Copernicus, Kepler, Galileo, and others into a new powerful synthesis.

It is remarkable that though the gravitational theory could not be verified at that time, there was hardly any doubt about its correctness. This is because Newton based his theory on sound scientific reasoning and backed it with mathematics. This made the theory simple and elegant. These qualities are now recognised as essential requirements of a good scientific theory.

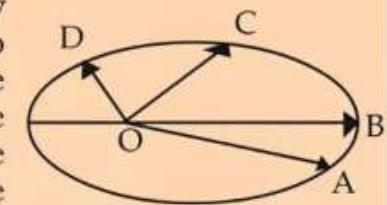
How did Newton guess the inverse-square rule?

There has always been a great interest in the motion of planets. By the 16th century, a lot of data on the motion of planets had been collected by many astronomers. Based on these data Johannes Kepler derived three laws, which govern the motion of planets. These are called Kepler's laws. These are:

1. The orbit of a planet is an ellipse with the Sun at one of the foci, as shown in the figure given below. In this figure O is the position of the Sun.
2. The line joining the planet and the Sun sweep equal areas in equal intervals of time. Thus, if the time of travel from A to B is the same as that from C to D, then the areas OAB and OCD are equal.
3. The cube of the mean distance of a planet from the Sun is proportional to the square of its orbital period T . Or, $r^3/T^2 = \text{constant}$.

It is important to note that Kepler could not give a theory to explain the motion of planets. It was Newton who showed that the cause of the planetary motion is the gravitational force that the Sun exerts on them. Newton used the third law

of Kepler to calculate the gravitational force of attraction. The gravitational force of the earth is



weakened by distance. A simple argument goes like this. We can assume that the planetary orbits are circular. Suppose the orbital velocity is v and the radius of the orbit is r . Then the force acting on an orbiting planet is given by $F \propto v^2/r$.

If T denotes the period, then $v = 2\pi r/T$, so that $v^2 \propto r^2/T^2$.

We can rewrite this as $v^2 \propto (1/r) \times (r^3/T^2)$. Since r^3/T^2 is constant by Kepler's third law, we have $v^2 \propto 1/r$. Combining this with $F \propto v^2/r$, we get, $F \propto 1/r^2$.

lecture . no. 7

Differentiation

• $y = 2x$

x	1	2	3	4
y	2	4	6	8

$$\frac{dy}{dx}$$

$$\frac{\emptyset}{\emptyset}$$

differentiation = $\frac{\text{Small change in } y}{\text{Small change in } x}$

Small change

$$= \frac{(y_2 - y_1)}{(x_2 - x_1)} = \frac{dy}{dx}$$

Small change

lecture . no. 7

Differentiation

• $y = 2x$

x	1	2	3	4
y	2	4	6	8

$x_1 \rightarrow 1, x_2 \rightarrow 2, x_3 \rightarrow 3, x_4 \rightarrow 4$
 $y_1 \rightarrow 2, y_2 \rightarrow 4, y_3 \rightarrow 6, y_4 \rightarrow 8$

~~$\frac{dy}{dx}$~~

~~$\frac{\emptyset}{\emptyset}$~~

$y = 2x$

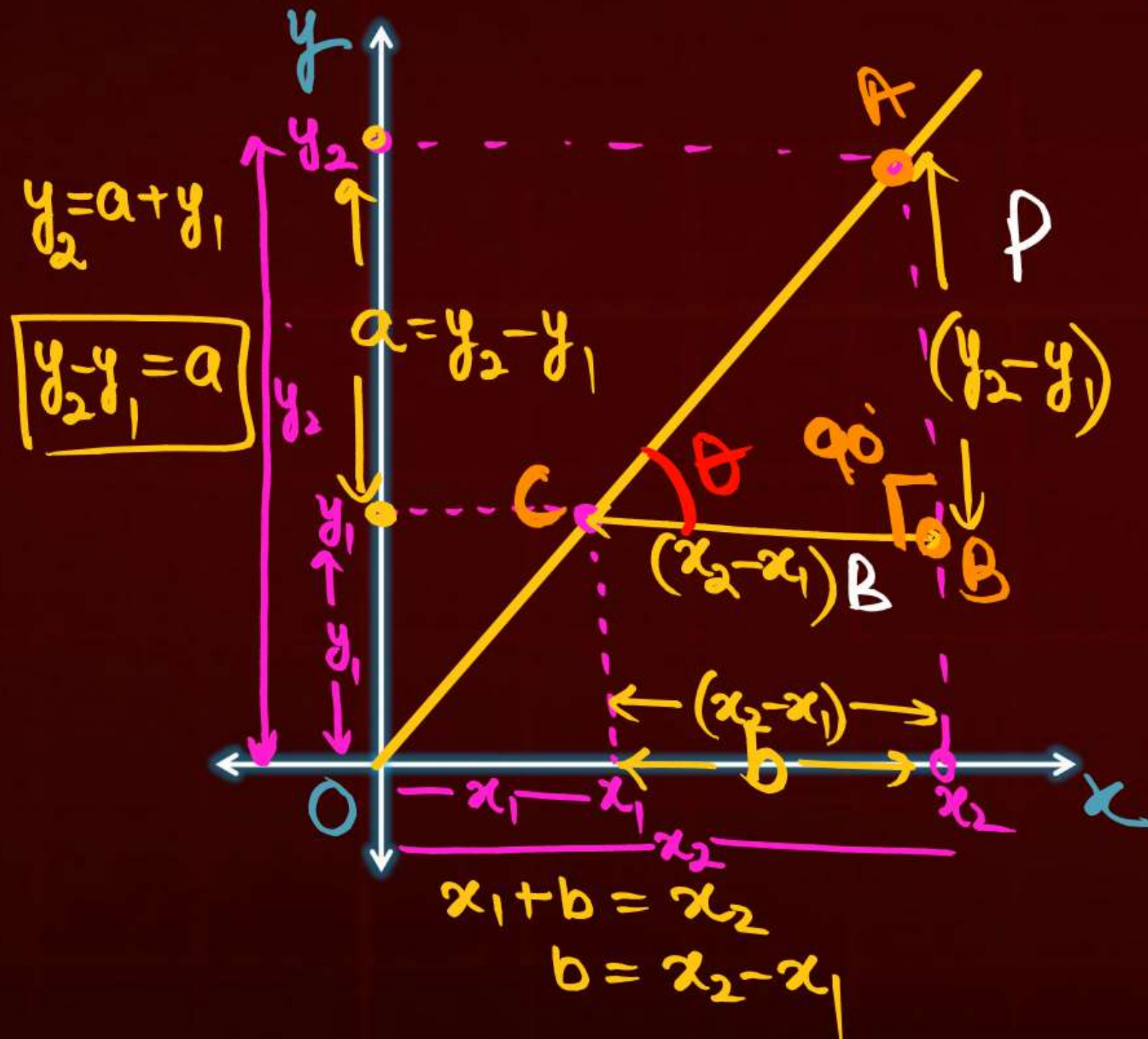
$m(\text{slope}) / \tan \theta$

$\frac{dy}{dx}$

differentiation = $\frac{dy}{dx} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{4 - 2}{2 - 1} = \frac{2}{1} = 2$

$\frac{dy}{dx} = \text{Slope}(m) = \tan \theta \checkmark$

Differentiation



In rt ΔABC

$$\tan \theta = \frac{P}{B} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{dy}{dx}$$

$\tan \theta = \frac{dy}{dx}$

Slope(m)

Differentiation

Rule-1

$$\boxed{y = c} \quad \frac{dy}{dx} = ?$$

Sol $\frac{dy}{dx} = \frac{dc}{dx} = 0$

Proof

$$\frac{dy}{dx} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{2 - 2}{x_2 - x_1} = \frac{0}{x_2 - x_1} = 0$$

Q $y = 2$

Sol $\frac{dy}{dx} = \frac{d2}{dx} = 0$

Q $y = 5$

Sol $\frac{dy}{dx} = \frac{d5}{dx} = 0 \checkmark$

Differentiation

Rule-2

$$y = x^n$$

$$\frac{dy}{dx} = \frac{dx^n}{dx} = \underline{\underline{nx^{n-1}}}$$

Q $y = x^3$

Sol $\frac{dy}{dx} = \frac{dx^3}{dx} = 3x^{3-1} = \underline{3x^2}$

Q $y = x^4$

Sol $\frac{dy}{dx} = \frac{dx^4}{dx} = 4x^{4-1} = \underline{4x^3}$

Q $y = x^{10}$

$\frac{dy}{dx} = \frac{dx^{10}}{dx} = 10x^{10-1} = \underline{10x^9}$

h/w

$$\underline{y = x^{100}}$$

$$\frac{dy}{dx} = ?$$

Differentiation

Rule-3

$$y = e^x \text{ (Jee-Basic Numerical)}$$

$$\frac{dy}{dx} = \frac{de^x}{dx} = e^x$$

$$\text{Q} \quad y = e^x.$$

$$\text{Sol} \quad \frac{dy}{dx}$$

$$\frac{de^x}{dx} = \underline{e^x} \checkmark$$

Differentiation

Rule-4

$$y = \ln x$$

$$\frac{dy}{dx} = \frac{d}{dx} \ln x = \frac{1}{x}$$

$$\underline{\underline{Q}} \quad y = \ln x$$

$$\underline{\underline{Sol}} \quad \frac{dy}{dx} = \frac{d \ln x}{dx} = \frac{1}{x}$$

$$\underline{\underline{4/10}} \quad \underline{\underline{Q}} \quad y = \underline{\underline{\ln x}}$$

Differentiation

Rule-5

$y = \sin x$ (Neet-Basic Numerical)

$$\frac{dy}{dx} = \frac{d\sin x}{dx} = \underline{\cos x}$$

Q $y = \sin x$

Sol $\frac{dy}{dx} = \frac{d\sin x}{dx} = \underline{\cos x}$ ✓

Differentiation

Rule-6

$y = \cos x$ (Neet - Basic Numerical use)

$$\frac{dy}{dx} = \frac{d(\cos x)}{dx} = -\underline{\sin x}$$

Q $y = \cos x$

Sol $\frac{dy}{dx} = \frac{d(\cos x)}{dx}$

$= -\underline{\sin x}$

Differentiation

Rule-7

$$y = \underline{\tan x}$$

$$\frac{dy}{dx} = \frac{d \tan x}{dx} = \underline{\underline{\sec^2 x}}$$

$$\underline{\underline{Q}} \quad \frac{dy}{dx}$$

$$\underline{\underline{Sol}} \quad \frac{d \tan x}{dx} = \underline{\sec^2 x}$$

$$\underline{\underline{Q}} \quad \frac{dy}{dx}$$

$$\underline{\underline{Sol}} \quad \frac{d \tan x}{dx} = \underline{\sec^2 x}$$

Differentiation

Rule-8

$$y = \operatorname{cosec} x$$

$$\frac{dy}{dx} = \frac{d(\operatorname{cosec} x)}{dx} = \underline{-\operatorname{cosec} x \cot x}$$

Q $y = \operatorname{cosec} x$.

Sol $\frac{dy}{dx} = \frac{d(\operatorname{cosec} x)}{dx}$

$$= \underline{-\operatorname{cosec} x \cot x}$$

Q $y = \operatorname{cosec} x$

Sol $\frac{dy}{dx} = \frac{d(\operatorname{cosec} x)}{dx} = -\operatorname{cosec} x \cot x$

Differentiation

Rule-9

$$y = \sec x$$

$$\frac{dy}{dx} = \frac{d \sec x}{dx} = \underline{\sec x \tan x}$$

Q $y = \sec x$.

Sol $\frac{dy}{dx}$

$$\frac{d \sec x}{dx} = \underline{\sec x \tan x}$$

Differentiation

Rule-10

$$y = \underline{\cot x}$$

$$\frac{dy}{dx} = \frac{d \cot x}{dx} = - \underline{\operatorname{cosec}^2 x}$$

$$y = \underline{\cot x} \checkmark$$

$$\frac{dy}{dx} = \frac{d \cot x}{dx}$$

$$= - \underline{\operatorname{cosec}^2 x} \checkmark$$

Differentiation

Rule-11

Sum Rule

$$y = u + v + w \quad (\text{Neet})$$

$$\frac{dy}{dx} = \frac{d}{dx}(u + v + w)$$

$$= \frac{du}{dx} + \frac{dv}{dx} + \frac{dw}{dx}$$

Q $y = x^3 + x^2 + \sin x$

$$\frac{dy}{dx} = ?$$

Sol $\frac{dy}{dx} = \frac{d}{dx}(x^3 + x^2 + \sin x)$

$$= \frac{dx^3}{dx} + \frac{dx^2}{dx} + \frac{d\sin x}{dx}$$

$$= 3x^{3-1} + 2x^{2-1} + \cos x$$

$$= 3x^2 + 2x + \underline{\cos x}$$

Q

$$y = x^{10} + e^x + \cos x$$

(Neet)

2017 ✓

$$\frac{dy}{dx} = ?$$

Sol

$$\frac{dy}{dx}$$

$$= \frac{d}{dx} (x^{10} + e^x + \cos x)$$

$$= \frac{dx^{10}}{dx} + \frac{de^x}{dx} + \frac{d(\cos x)}{dx}$$

$$= 10x^{10-1} + e^x + -\sin x$$

$$= 10x^9 + e^x - \underline{\underline{\sin x}}$$

Differentiation

Rule-12

Subtraction Rule.

$$y = u - v - w$$

$$\frac{dy}{dx} = \frac{d}{dx}(u - v - w)$$

$$= \frac{du}{dx} - \frac{dv}{dx} - \frac{dw}{dx}$$

Q $y = x^3 - x^4 - \sin x$

Sol $\frac{dy}{dx} = \frac{d}{dx}(x^3 - x^4 - \sin x)$

$$= \frac{dx^3}{dx} - \frac{dx^4}{dx} - \frac{d\sin x}{dx}$$

$$= 3x^{3-1} - 4x^{4-1} - \cos x$$

$$= 3x^2 - 4x^3 - \cos x$$

Differentiation

Rule-13

→ Sum(+) and Substraction)

$$y = u + v - w$$

$$\frac{dy}{dx} = \frac{d}{dx}(u + v - w)$$

$$= \frac{du}{dx} + \frac{dv}{dx} - \frac{dw}{dx}$$

Q $y = x^5 + x^3 - \tan x$

$$\frac{dy}{dx} = ?$$

Sol $\frac{dy}{dx} = \frac{d}{dx}(x^5 + x^3 - \tan x)$

$$= \frac{dx^5}{dx} + \frac{dx^3}{dx} - \frac{d \tan x}{dx}$$

$$= 5x^{5-1} + 3x^{3-1} - \sec^2 x$$

$$= 5x^4 + 3x^2 - \sec^2 x$$

Rule no. 14.

(Multiplication Rule)

$$y = u \cdot v \quad (\text{UV-Rule})$$

$$\frac{dy}{dx} = \frac{d(uv)}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

Q $y = \underbrace{x^3}_u \cdot \underbrace{\sin x}_v$

Sol $\frac{dy}{dx} = \frac{d}{dx} \underbrace{x^3}_u \cdot \underbrace{\sin x}_v$

$$= \underbrace{x^3}_u \frac{d \underbrace{\sin x}_v}{dx} + \underbrace{\sin x}_v \frac{d \underbrace{x^3}_u}{dx}$$

$$= x^3 \cdot \cos x + \sin x \cdot 3x^{3-1}$$

$$= x^3 \cos x + \sin x \cdot 3x^2$$



Q $y = x^4 \cdot \tan x$
 $\underbrace{x^4}_u \cdot \underbrace{\tan x}_v$

$\frac{d}{dx} (u \cdot v) = \frac{duv}{dx} = \underbrace{x^4}_u \frac{d \tan x}{dx} + \underbrace{\tan x}_v \frac{d x^4}{dx}$

$$= x^4 \frac{d \tan x}{dx} + \tan x \frac{d x^4}{dx}$$

$$= x^4 \sec^2 x + \tan x \cdot 4x^{4-1}$$

$$= x^4 \sec^2 x + \tan x \cdot 4x^3$$

u/w ✓

$$y = x^3 \cdot \cos x$$

Rule. no. 15 (Division Rule)

$$y = \frac{u}{v}$$

jee

$$\frac{dy}{dx} = \frac{d\left(\frac{u}{v}\right)}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$\frac{dy}{dx} = \frac{d\frac{u}{v}}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

Q $y = \tan x$ $\frac{dy}{dx} = ?$

$\frac{dy}{dx} = \frac{d \tan x}{dx} = \sec^2 x$

Sol $y = \tan x$ • $\tan x = \frac{\sin x}{\cos x}$

$y = \left(\frac{\sin x}{\cos x} \right)^u$ (Neet-level Practice)

$$\frac{dy}{dx} = \frac{d\left(\frac{u}{v}\right)}{dx} = \frac{\overset{\cos x}{\underbrace{v}} \frac{du}{dx} - \underbrace{u}_{\sin x} \frac{dv}{dx}}{v^2 \rightarrow \cos^2 x}$$

$$= \frac{\cos x \frac{d \sin x}{dx} - \sin x \frac{d \cos x}{dx}}{(\cos x)^2}$$

$$= \frac{\cos x \cdot \cos x - \sin x \cdot -\sin x}{(\cos x)^2}$$

$$= \frac{\cos^2 x + -\sin x \cdot -\sin x}{(\cos x)^2}$$

$$= \frac{\cos^2 x + \sin^2 x}{(\cos x)^2}$$

$$= \frac{(\sin^2 x + \cos^2 x)}{(\cos x)^2} \quad \begin{matrix} \sin^2 \theta + \cos^2 \theta \\ = 1 \end{matrix}$$

$$= \frac{1}{(\cos x)^2} = \frac{1}{\cos^2 x}$$

$$= \frac{1}{\cos^2 x}$$

$$= \underline{\sec^2 x}$$

$$\# \quad \frac{1}{\cos x} = \sec x$$

$$\left(\frac{1}{\cos x} \right)^2 = (\sec x)^2$$

$$\frac{1^2}{(\cos x)^2} = \sec^2 x$$

$$\frac{1}{\cos^2 x} = \sec^2 x$$

Rule no. 16

$$y = cu$$

$$\frac{dy}{dx} = \frac{d(cu)}{dx} = c \frac{du}{dx}$$

$$y = 2x^3 \quad (2022) \\ \text{JKROSE})$$

$$\frac{dy}{dx} = \frac{d 2x^3}{dx}$$

$$= 2 \frac{dx^3}{dx}$$

$$= 2 \cdot 3x^{3-1}$$

$$= \underline{\underline{6x^2}}$$

Q10 $y = 5x^4$

$\frac{dy}{dx} = ?$

Jee-level

Plzz Yaha tak

Complete

Karna

QUESTION [JKCET 2012]

$\frac{dy}{dx}$ is?

- 1 Area of Graph
- 2 $\frac{y_2 - y_1}{x_2 - x_1}$
- 3 Slope of the graph
- 4 Both 2 and 3 ✓

SUMMARY



Differentiation ✓
Rules ✓

QUESTION [NEET 2019]

$$y = \tan x \quad (\text{Numerical})$$

$$\frac{dy}{dx} = \frac{d \tan x}{dx}$$

Thank
You